



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

RentWise: Simplified Property Management Portal

The domain of the Project:
FULL STACK DEVELOPMENT-G29

Team Mentors (and their designation):
PRASANNA KUNDU
(SURE TRUST MENTOR)

Team Member:
Ms. SRENI SIVADAS

January 2026 to March 2026



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Declaration

The project titled “**RentWise: Simplified Property Management Portal**” has been mentored by **Prasanna Kundu**, organised by SURE Trust, from August 2025 to February 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:

Ms. SRENI SIVADAS

Mentor's Name

PRASANNA KUNDU—Sure Trust Mentor

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

RentWise is a web-based property management portal designed to simplify the interaction between landlords and tenants. The system provides a centralized platform for managing rental properties, tracking rent payments, handling maintenance requests, and enabling real-time communication.

The primary objective of this project is to eliminate inefficiencies in traditional property management systems, which often rely on manual tracking and fragmented communication. RentWise addresses these challenges by offering structured dashboards for both landlords and tenants.

Landlords can manage multiple properties, monitor tenant details, and track rent payment statuses, while tenants can view lease information, make rent payments (mock system), and submit maintenance requests with supporting images or documents. The system also includes a real-time messaging feature to improve communication and transparency.

The project is developed using modern web technologies, including a RESTful API architecture, database management systems, and real-time communication using WebSockets. These technologies ensure scalability, efficiency, and a user-friendly experience.

Overall, RentWise demonstrates a practical and industry-relevant solution that models real-world scenarios such as rent management and maintenance workflows. It provides valuable hands-on experience in full-stack development and can be further enhanced with features like online payment integration and mobile support.



Introduction

The rapid growth of digital solutions has transformed how everyday operations are managed across industries, including real estate and property management. Traditionally, landlords and tenants rely on manual methods such as spreadsheets, phone calls, and messaging apps to handle rent payments, maintenance requests, and communication. These methods often lead to inefficiencies, lack of transparency, and difficulty in tracking important information.

The primary problem addressed by this project is the absence of a centralized system that allows landlords and tenants to efficiently manage rental activities. Landlords face challenges in tracking multiple properties, monitoring rent payments, and handling maintenance issues, while tenants often experience delays in communication and request resolution.

RentWise is designed to solve these problems by providing a unified web-based platform that streamlines property management processes. The system enables landlords to manage properties, tenants, leases, and maintenance requests in one place, while tenants can easily access their rental information, pay rent (mock functionality), and raise maintenance requests.

The scope of the project is focused on a single property or a small building setup, making it suitable for small-scale landlords. The system includes essential features such as dashboards, payment tracking, maintenance request management, and messaging. However, it has certain limitations, such as the absence of real payment gateway integration and advanced analytics features.

An important innovation in this project is the integration of multiple functionalities—property management, maintenance tracking, and real-time communication—into a single platform. Additionally, the use of modern technologies such as REST APIs and WebSockets enables efficient data handling and real-time updates, making the system more responsive and user-friendly.



Project Objectives

The main objective of the RentWise project is to design and develop a simplified property management system that streamlines interactions between landlords and tenants. The project aims to provide an efficient, user-friendly platform that reduces manual effort and improves transparency in managing rental activities.

One of the key goals is to create a centralized system where landlords can manage properties, tenants, leases, rent payments, and maintenance requests in an organized manner. At the same time, tenants should be able to access their rental details, view rent due dates, make rent payments (mock system), and submit maintenance requests بسهولة.

Another important objective is to implement a robust backend using RESTful APIs to handle data operations such as creating, updating, and retrieving records. The project also aims to design a well-structured database that maintains relationships between users, properties, leases, payments, and maintenance requests.

Additionally, the project focuses on enhancing communication between landlords and tenants through a real-time messaging system. This ensures faster issue resolution and better coordination.

The expected outcomes of this project include a fully functional web application with separate dashboards for landlords and tenants, a working maintenance request system with status tracking, a mock rent payment module, and a real-time chat feature. The final deliverable will be a complete full-stack application along with proper documentation and a deployable system that demonstrates practical implementation of modern web development technologies.



Methodology and Results

● **Methods/Technology used**

The RentWise project is developed using modern full-stack web development technologies. The frontend is built using HTML, CSS, and JavaScript (or React for dynamic UI). The backend is implemented using Node.js and Express to create RESTful APIs. The system uses a database such as MongoDB or MySQL to store and manage application data efficiently. Real-time communication is enabled using WebSockets for instant messaging between users.

● **Tools / Software Used**

The following tools were used during development:

- **IDE:** Visual Studio Code
- **Version Control:** Git & GitHub
- **API Testing:** Postman
- **Database Tool:** MongoDB Compass
- **Package Manager:** npm (Node Package Manager)
- **Browser Tools:** Chrome Developer Tools (debugging & testing)

● **Data Collection Approach**

The project uses **mock and system-generated data** to simulate real-world scenarios.

- User data (landlord/tenant) is entered manually during registration
- Lease details, rent payments, and maintenance requests are created within the system
- File uploads (images/videos) simulate real maintenance issues
- No external datasets are required

● **Project Architecture**

The system follows a **3-Tier Architecture**:

1. **Frontend (Client Layer)**
 1. Built using React.js
 2. Handles user interaction and UI
2. **Backend (Application Layer)**



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1. Node.js + Express.js
2. Processes business logic and API requests

3. Database (Data Layer)

1. MongoDB
2. Stores users, properties, leases, payments, and requests

Communication Flow:

Frontend → REST API → Backend → Database

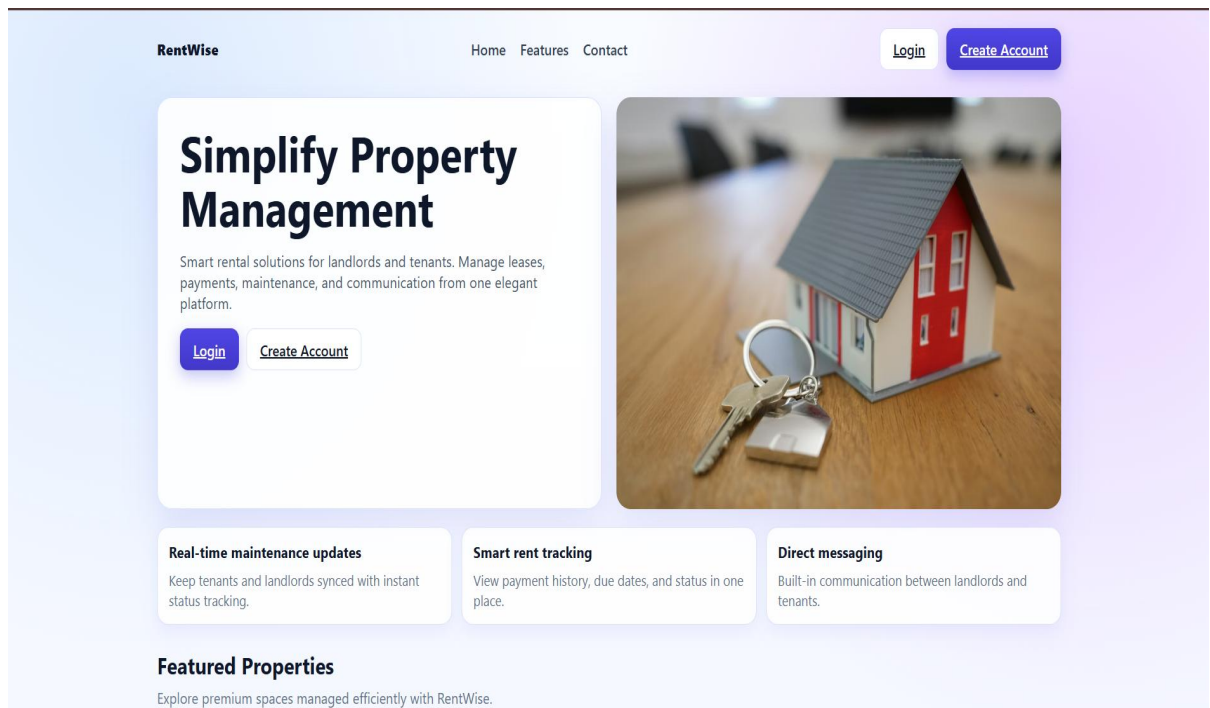
Real-Time Feature:

Socket.io enables live messaging and instant status updates

● **Project Results (Working System)**

The RentWise system was successfully implemented with the following features:

- Landlord dashboard to manage properties and tenants
- Tenant dashboard to view rent and submit requests
- Rent payment tracking system (mock payments)
- Maintenance request module with status updates
- Real-time chat between landlord and tenant





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RentWise

Dashboard

Messaging

Logout

Messaging

Chat Room - Lease 09f3ea

09f3ea - \$10000

Messaging with Tenant for this lease.

i have paid the rent
4/25/2026, 8:13:15 PM

hii
4/25/2026, 8:10:18 PM

Type a message...

Send

RentWise

Dashboard

Properties

Tenants

Payments

Requests

Profile

Logout

Landlord Dashboard

Overview your portfolio, tenants, collections and requests.

Notifications

Sivadas

Total Properties

3

Active Tenants

3

Monthly Revenue

\$80,000

Pending Requests

0

Add Property

Address

Add Property

Create Lease

Property

chennai

Tenant

test (test@gmail.com)

Start

dd-mm-yyyy

End

dd-mm-yyyy

Rent

Lease PDF

Choose File No file chosen



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RentWise

- Dashboard
- My Rentals
- Payments
- Maintenance
- Profile
- Logout

Tenant Dashboard

Manage your rentals, payments, and maintenance with ease.

Current Rent
\$10000

Due Date
5/8/2026

Payment Status
successful

Active Requests
0

My Rentals

chennai
Lease: 4/8/2026 - 7/20/2026
Rent: \$10000

Chennai
Lease: 2/18/2026 - 11/23/2027
Rent: \$20000

Payment History

Pay Rent

Amount	Status	Date
\$10000	successful	4/25/2026, 8:11:08 PM

- GitHub Link

Frontend Repo Link: <https://github.com/srenisivadas/rentwise-frontend>

Backend Repo Link : <https://github.com/srenisivadas/rentwise-backend>

Deployed Link : <https://rentwise-frontend.onrender.com>



Learning and Reflection

This project provided valuable hands-on experience in full-stack web development and helped in understanding how real-world applications are built and managed. During the development of the RentWise system, I gained practical knowledge in both frontend and backend technologies.

From a technical perspective, I learned how to build dynamic user interfaces using React.js and how to create RESTful APIs using Node.js and Express. I also developed a strong understanding of database design and management using MongoDB, including creating schemas and managing relationships between different entities such as users, properties, and maintenance requests. Additionally, I explored real-time communication using Socket.io, which enabled live messaging functionality within the application.

I also improved my skills in using development tools such as Git and GitHub for version control, Postman for API testing, and Visual Studio Code for efficient coding and debugging. Working on this project enhanced my problem-solving abilities, especially in debugging errors and integrating different components of the system.

From an overall experience perspective, this project helped me understand the complete lifecycle of application development—from planning and designing the architecture to implementation and testing. It also improved my ability to work independently, manage time effectively, and handle challenges during development.

Overall, the RentWise project was a highly beneficial learning experience that strengthened my technical skills and prepared me for real-world software development tasks.



Conclusion and Future Scope

The RentWise project was successfully designed and developed to simplify property management for landlords and tenants. The primary objective of creating a centralized platform to manage properties, track rent payments, handle maintenance requests, and enable communication was achieved. The system demonstrates the effective use of full-stack technologies, including React for the frontend, Node.js and Express for the backend, and MongoDB for data management.

Through this project, a functional application was built with key features such as landlord and tenant dashboards, rent tracking (mock payments), maintenance request management, file uploads, and real-time messaging. These features closely simulate real-world property management scenarios and provide a practical solution to common challenges faced by users.

In terms of future scope, the project can be enhanced by integrating a secure online payment gateway to enable real rent transactions. A mobile application version can also be developed to improve accessibility and user experience. Additional features such as email and SMS notifications, advanced analytics for rent tracking, and role-based access control can further improve the system. The platform can also be scaled to support multiple properties and larger real estate businesses.

Overall, RentWise serves as a strong foundation for a scalable and industry-relevant application and reflects the practical implementation of modern web development concepts.